bio

12 · CBSE

Based on 4 year(s) of previous papers

Topic Priority Overview

Topic	Priority	Confidence	Freq	Marks
Microspore formation: meiosis of pollen mother cell	High	High	3×	1-3 marks (MCQ / short)
Ploidy of primary and secondary spermatocytes (spermatogenesis)	High	High	3×	2 marks (short answer)
Trophoblast function in human embryo	Medium	Medium	2×	2 marks (short)
Polygenic inheritance: deviation from Mendelian patterns	High	High	3×	2-3 marks
Non-Mendelian inheritance in human blood groups (multiple allelism & co-dominance)	High	High	3×	2-3 marks
Tears as immune barrier: type and example	Medium	Medium	2×	2 marks
Methods to introduce alien DNA into plant and animal cells (e.g., microinjection, Agrobacterium)	High	High	3×	2-5 marks depending on depth
Logistic population growth model (Verhulst-Pearl) and equation (K, r)	High	High	3×	2-5 marks
Decomposition pathway to humus (stages: fragmentation, catabolism, mineralisation, humification)	Medium	Medium	2×	2-3 marks
Wind-pollinated plant adaptations (3 adaptations)	Medium	Medium	2×	3 marks
Uterus wall structure: layers and functions (endometrium, myometrium, perimetrium)	High	High	3×	3-5 marks
Replication fork: definition and role in replicating long DNA strands	Medium	Medium	2×	3 marks
DNA polymorphism: definition and two applications	Medium	Medium	2×	3 marks
Baculoviruses used as biocontrol: three reasons	Low	Low	1×	3 marks
Pathogen detection at low concentrations: PCR and diseases detected (two examples)	Medium	Medium	2×	3 marks

Predator vs Prey: distinction and two ecological roles of predators	High	High	3×	3 marks
Hardy-Weinberg equilibrium: definition and factors (founder effect, gene migration)	High	High	3×	4 marks (case-based)
Mutation as a source of evolution (how mutations result in evolution)	Medium	Medium	2×	2-4 marks (case option)
Long-term immunity from vaccination: name, property (memory), types and memory cells/ anamnestic response	High	High	3×	4 marks (case-based)
Angiosperm seed advantages and features: seed vs perisperm vs pericarp and polyembryony example	High	High	3×	5 marks (long answer)
Gonadotropin source and ovarian changes during menstrual cycle	Medium	Medium	2×	5 marks (long subpart)
Placenta-only hormones produced during pregnancy (name two)	Medium	Medium	2×	5 marks subpart (3 short items total)
Stem cells in human embryo: location and significance	Medium	Medium	2×	5 marks subpart
Insulin structure, maturation and immune reaction to animal insulin	Low	Low	1×	5 marks (long option)
Mutualistic interaction and co-evolution example (Ophrys-bee type)	Medium	Medium	2×	3-5 marks
Endemism definition and hotspots in India (name two)	Medium	Medium	2×	5 marks subpart
Population density calculation from natality, mortality, immigration, emigration (basic arithmetic problem)	Medium	Medium	2×	5 marks (calculation)
Activated sludge in Sewage Treatment Plant (STP) and anaerobic sludge digestion	High	High	3×	3-5 marks
Recombinant DNA technology steps: cloning, transfer, replication and term 'replication of recombinant DNA' (technical term for Step 4)	High	High	3×	2-5 marks
Restriction endonuclease naming convention and recognition site length (e.g., EcoRI, HindII, 4/6 base cutters)	High	High	3×	1-3 marks (MCQ/short)
Spermatogenesis hormones: site of FSH action, LH role, and accessory ducts path	Medium	Medium	2×	2-4 marks
Competence of bacterial cell for transformation using calcium ions	Low	Low	1×	2-3 marks
Ribozyme identity (23S rRNA) and catalytic rRNA	Medium	High	1×	1 mark (MCQ)

Assisted Reproductive Technologies (ART) steps (test-tube baby)	High	High	2×	2 marks (short) to 5 marks (long explanation)
Selection of transformed bacteria using antibiotic markers (ampicillin / tetracycline) and insertional inactivation	High	High	3×	1-3 marks (MCQ/ short), up to 4 marks in diagram/case

Detailed Study Notes

Read these notes carefully — they're based on actual patterns found in your past papers.

1. Microspore formation: meiosis of pollen mother cell

High Conf. High Priority

Appeared in: Paper 1 · Paper 3 · Paper 4 · 3× total

Question types: MCQ: 1-2

Kya Padhna Hai:

- Definition: 'Microspores are formed by' (pollen mother cell meiosis) as asked in MCQ
- Process step: Meiosis I and II in pollen mother cell producing tetrad of microspores
- Contrast: pollen microspore origin vs megaspore origin (contextual MCQ usage)
- Related term: 'pollen mother cell' nomenclature used verbatim in MCQ
- Application: identification in MCQ options (recognize meiosis vs mitosis answers)

Kaise Poochha Jaata Hai:

Mostly MCQ: exact phrasing 'Microspores are formed by' with options listing meiosis/mitosis of pollen mother cell/polar nuclei/tapetal cells

Repeat Pattern:

Similar direct MCQ asked across papers (Paper 1 MCQ, Paper 3 MCQ, Paper 4 MCQ) — high repeatability for single-line factual recall

Key Terms:

microspore

pollen mother cell

meiosis

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2. Ploidy of primary and secondary spermatocytes (spermatogenesis)

High Conf. High Priority

Appeared in: Paper 1 · Paper 3 · Paper 4 · 3× total

Question types: Short: 2-3

Kya Padhna Hai:

- Ploidy values: primary spermatocyte = diploid (2n), secondary spermatocyte = haploid (n) as asked verbatim
- Related cell transitions: spermatogonia !' primary spermatocyte (mitosis to meiosis I entry)
- Named processes: first meiotic division differences (compare with oogenesis) which appeared as linked short question
- Accessory: site/location (seminiferous tubules) sometimes tied to spermatogenesis questions

Kaise Poochha Jaata Hai:

Very short answer: 'What is the ploidy of primary spermatocytes and secondary spermatocytes ?' or as part of spermatogenesis case question

Repeat Pattern:

Phrased similarly in Paper 1 and Paper 3 and appears in a Paper 4 long-case about spermatogenesis — consistent short-answer recall

Key Terms:

- primary spermatocyte
- secondary spermatocyte
- ploidy

3. Polygenic inheritance: deviation from Mendelian patterns

High Conf. High Priority

Appeared in: Paper 1 · Paper 2 · Paper 3 · 3× total

Question types: Short: 2-3

Kya Padhna Hai:

- Exact phrasing: 'How does polygenic inheritance show deviation from Mendelian inheritance ?'
- Features: multiple genes contribute, continuous variation, non-Mendelian phenotypic ratios as expected in questions
- Example linkage: skin colour or height often implied when explaining deviation
- Comparison bullet: contrast with single-gene Mendelian inheritance (F1/F2 ratios not applicable)

Kaise Poochha Jaata Hai:

Short answer (2 marks) requiring explanation of deviation with one or two sentences and example

Repeat Pattern:

Directly asked in Paper 1; related multiple-allelic/co-dominance questions in Paper 3 and Paper 2 — concept recurs

Key Terms:

- polygenic inheritance
- continuous variation
- non-Mendelian

4. Non-Mendelian inheritance in human blood groups (multiple allelism & co-dominance)

High Conf. High Priority

Appeared in: Paper 1 · Paper 2 · Paper 3 · 3× total

Question types: Short: 2-3 | Long: 3 (in some papers)

Kya Padhna Hai:

- Explicit ask: 'Explain the two non-Mendelian patterns of inheritance that are observed in blood groups of humans.'
- Named mechanisms: 'multiple allelism' and 'co-dominance' — define and give ABO example
- Punnett/workout style: genotype/phenotype distinctions for ABO group appearing in long/short Qs
- Related calculation: inheritance examples and phenotypic ratios asked in Paper 3 and Paper 4 (justify co-dominance)

Kaise Poochha Jaata Hai:

Short explanatory question OR problem requiring reasoned justification that ABO shows multiple alleles and co-dominance; sometimes appears as a 3-mark justification

Repeat Pattern:

Appears across multiple papers (Paper 1 B, Paper 3 longer questions on ABO), strong repeat pattern

Key Terms:

- *ABO*
- *multiple alleles*
- *co-dominance*

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5. Methods to introduce alien DNA into plant and animal cells (e.g., microinjection, Agrobacterium)

High Conf. | High Priority

Appeared in: Paper 1 · Paper 2 · Paper 4 · 3× total

Question types: Short: 2-3 | Long: 1-2 (Paper 2/4 long answers)

Kya Padhna Hai:

- Direct phrasing: 'Name and explain one method which is used to introduce alien DNA into an animal cell and plant cell respectively.'
- Animal method named in papers: microinjection/electroporation or gene gun (when asked choose one)
- Plant method named in papers: Agrobacterium-mediated transformation or gene gun
- Technical detail: vector/Agrobacterium T-DNA delivery and microinjection basic mechanism (step-level) as required for 2-mark answer
- Alternative specifics: transformation competence using calcium ions for bacterial uptake (related competence question in Paper 3/2)

Kaise Poochha Jaata Hai:

Short 2-mark question expecting naming of one method per kingdom and 1-2 sentence explanation; sometimes appears in Section B or E

Repeat Pattern:

Consistently asked across papers as a paired plant/animal short Q — high repeat

Key Terms:

- *microinjection*
- *Agrobacterium*
- *electroporation*

6. Logistic population growth model (Verhulst-Pearl) and equation (K, r)

High Conf. High Priority

Appeared in: Paper 1 · Paper 2 · Paper 4 · 3× total

Question types: Short: 1-2 | Long: 1 (Paper 2/4 long question options)

Kya Padhna Hai:

- Exact wording: 'Describe the population logistic growth model and provide its equation.'
- Equation: Verhulst-Pearl logistic equation S = CAZ or the K-based form $dN/dt = rN(1 - N/K)$ — both forms referenced across papers
- Parameters: meaning of K (carrying capacity), r (intrinsic rate), and N (population density)
- Graph: logistic (sigmoid) curve features – lag, exponential, deceleration, stationary phases (asked in Paper 2 long answer)

Kaise Poochha Jaata Hai:

Short to long answer: either state equation + brief description (2 marks) or draw/explain logistic curve and equation (5 marks)

Repeat Pattern:

Appears in multiple papers in similar wording and in Section E long option — high reliability

Key Terms:

- Verhulst-Pearl
- carrying capacity
- $dN/dt = rN(1-N/K)$

7. Uterus wall structure: layers and functions (endometrium, myometrium, perimetrium)

High Conf. High Priority

Appeared in: Paper 1 · Paper 2 · Paper 4 · 3× total

Question types: Short: 3 | Long: 1-2

Kya Padhna Hai:

- Exact question: 'Describe the structure of the uterus wall and functions performed by any of its two layers.'
- Layers to define: endometrium (stratum functionalis/glandular secretions), myometrium (muscular contractions), perimetrium (serosa)
- Functional details: endometrium in implantation/menstrual shedding; myometrium in labour/expulsion
- Also appears alongside placenta/gonadotropin questions where uterine changes must be mapped to hormones

Kaise Poochha Jaata Hai:

Short 3-mark Q asking description + function of two named layers; sometimes appears in longer hormonal-context questions

Repeat Pattern:

Consistently asked across multiple papers as specific layer/function recall — high repeat

Key Terms:

- endometrium
- myometrium

perimetrium

8. Predator vs Prey: distinction and two ecological roles of predators

High Conf. High Priority

Appeared in: Paper 1 · Paper 2 · Paper 3 · 3× total

Question types: Short: 3

Kya Padhna Hai:

- Paper wording: 'Distinguish between predator and prey. Mention any two significant roles that predators play in nature.'
- Clear distinction: predator (kills/eats multiple prey) vs prey (is eaten) — include interaction examples
- Two roles: population control of prey, maintaining species diversity, removing weak/old individuals (specific roles requested)
- Example support: name an example predator-prey pair if asked (e.g., lion-zebra, praying mantis-caterpillar)

Kaise Poochha Jaata Hai:

Short 3-mark question: definition + two roles with brief explanation

Repeat Pattern:

Regularly appears in Section C across papers — high repeat

Key Terms:

predator

prey

population control

9. Hardy-Weinberg equilibrium: definition and factors (founder effect, gene migration)

High Conf. High Priority

Appeared in: Paper 1 · Paper 2 · Paper 3 · 3× total

Question types: Short: 1-2 | Case Study: 4 (case D question)

Kya Padhna Hai:

- Exact phrase: 'Define Hardy-Weinberg equilibrium.'
- Factors mentioned: gene migration, genetic drift, founder effect — define each (founder effect asked explicitly)
- Application: how migration changes allele frequencies and may lead to speciation (paper passage wording)
- Example answers: founders causing different allele frequency in new population (as in case passage)

Kaise Poochha Jaata Hai:

Case-based question with 3 subparts: definition, founder effect, and either gene migration effect or mutation causing evolution (internal choice present)

Repeat Pattern:

Case passage appears in Paper 1 and concept repeats in other papers' case questions — high repeat

Key Terms:

Hardy-Weinberg

founder effect

gene migration

10. Long-term immunity from vaccination: name, property (memory), types and memory cells/anamnestic response

High Conf. High Priority

Appeared in: Paper 1 · Paper 2 · Paper 3 · 3× total

Question types: Short: 1-2 | Long: 1 (case) | Case Study: 4

Kya Padhna Hai:

- Exact case phrasing: 'What name is given to such a type of immune response? Also name the property which is responsible for it.' (vaccination gives long protection)
- Name: 'immunological memory' or 'acquired active immunity' and property: 'memory cells'
- Types: humoral and cell-mediated immunity (question asks 'What are the types? Describe briefly.')
- Memory mechanics: role of memory B and T cells and anamnestic response (Paper 1 asked 'What is anamnestic response?')

Kaise Poochha Jaata Hai:

Case-based 4-mark question with (a) name+property, (b) types, (c) either memory cells role or definition of anamnestic response (internal choice)

Repeat Pattern:

Very similar case on vaccines/immunity appears across Papers 1-3 — high repeat

Key Terms:

immunological memory

memory B cells

anamnestic response

11. Angiosperm seed advantages and features: seed vs perisperm vs pericarp and polyembryony example

High Conf. High Priority

Appeared in: Paper 1 · Paper 2 · Paper 4 · 3× total

Question types: Short: 1-2 | Long: 5

Kya Padhna Hai:

- Exact Q: 'Describe how the presence of seed is advantageous to angiosperms. Mention any three points.'
- Specific distinctions required: 'Differentiate between perisperm and pericarp' phrasing present
- Polyembryony: definition and example required ('What is Polyembryony ? Give an example.')
- Seed structural terms: seed coat, embryo, endosperm functions as used in answer prompts

Kaise Poochha Jaata Hai:

Long 5-mark question with three subparts: seed advantages (3 points), perisperm vs pericarp differentiation, polyembryony definition+example

Repeat Pattern:

Seed advantage + perisperm/pericarp + polyembryony cluster recurs across papers (Paper 1 Section E, Paper 4 Section E) — high repeat

Key Terms:

- seed advantage
- perisperm
- pericarp

12. Activated sludge in Sewage Treatment Plant (STP) and anaerobic sludge digestion

High Conf. High Priority

Appeared in: Paper 2 · Paper 3 · Paper 4 · 3× total

Question types: Short: 3 | Long: 1-2

Kya Padhna Hai:

- Paper 2 direct: 'What do you mean by activated sludge in an STP? Explain the biological treatment of the major part of the sludge transferred from the aeration tank into the anaerobic sludge digesters.'
- Definition: activated sludge = biomass of aerobic microbes in aeration tank
- Anaerobic digestion steps: hydrolysis !’ acidogenesis !’ methanogenesis; methane/CO2 gases produced (papers mention gas mix)
- Purpose: reduce BOD and stabilise sludge before release; role of inoculum (small part of activated sludge) appears in MCQ

Kaise Poochha Jaata Hai:

Short 3-mark Q plus a 3-mark explanation of anaerobic sludge digestion; MCQs also test 'inoculum in STP' phrasing

Repeat Pattern:

Core STP theme appears across Papers 2,3,4 including MCQs and long subparts — high repeat

Key Terms:

- activated sludge
- anaerobic digester
- methanogenesis

13. Recombinant DNA technology steps: cloning, transfer, replication and term 'replication of recombinant DNA' (technical term for Step 4)

High Conf. High Priority

Appeared in: Paper 2 · Paper 3 · Paper 4 · 3× total

Question types: Short: 2-3 | Long: 1-2

Kya Padhna Hai:

- Paper text: flow diagram Step 1 (cut plasmid + alien DNA), Step 2 (recombinant DNA), Step 3 (transfer into E. coli), Step 4 'Replication of the recombinant DNA molecule in E. coli to form multiple copies of the alien gene' — name Step 4 technical term (cloning/amplification)
- Technical term expected: 'cloning' or 'replication to form multiple copies' sometimes called 'amplification' or 'clonal propagation of vector'
- Complementary MCQ/justification: choice of restriction enzymes (EcoRI + HindIII vs both EcoRI) to produce compatible ends (Paper 2 question)

- Vector sites: rop/pBR322 cloning site and antibiotic marker mapping appear in MCQs

Kaise Poochha Jaata Hai:

Short answer asking technical term (Step 4) and justification of enzyme choice; MCQ on cloning site and enzyme identity also used

Repeat Pattern:

Recombinant DNA flow diagram and enzyme selection repeatedly tested across Paper 2 and Paper 3 — high repeat

Key Terms:

- *recombinant DNA*
- *cloning*
- *EcoRI*

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14. Restriction endonuclease naming convention and recognition site length (e.g., EcoRI, HindII, 4/6 base cutters)

High Conf. | High Priority

Appeared in: Paper 2 · Paper 3 · Paper 4 · 3× total

Question types: MCQ: 2-3 | Short: 1-2

Kya Padhna Hai:

- Paper phrases: 'How is a restriction endonuclease named? Explain with example.' and MCQ: 'Hind II recognizes six base pairs/four base pairs?'
- Naming rule: genus-species-strain- enzyme number convention (e.g., EcoRI = Escherichia coli RY13)
- Recognition site length examples: HindII/EcoRI often cited as 6 bp cutters; MCQs asked to choose correct bp length
- Cloning implications: compatible sticky/blunt ends when choosing enzymes for plasmid/insert (Paper 2 enzyme-choice question)

Kaise Poochha Jaata Hai:

Short 2-mark explanation for naming + MCQ recognition length selection; enzyme combination justification in recombinant flow diagram

Repeat Pattern:

Appears multiple times across papers (Paper 2 enzyme choice, Paper 3 naming, Paper 4 EcoRI convention) — high repeat

Key Terms:

- *EcoRI*
- *HindII*
- *recognition site*

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15. Assisted Reproductive Technologies (ART) steps (test-tube baby)

High Conf. High Priority

Appeared in: Paper 4 · Paper 3 · 2× total

Question types: MCQ: 0 | Short: 1-2 | Long: 1 | Case Study: 0

Kya Padhna Hai:

- Core steps to memorise: ovarian stimulation and monitoring !’ oocyte retrieval !’ in vitro fertilization (mixing sperm + oocyte in lab) !’ embryo culture !’ embryo transfer into uterus (and optionally ICSI/GIFT/ZIFT alternatives). Know the brief purpose of each step.
- Why called 'test-tube baby': embryo fertilisation and early development occur in vitro (in laboratory glassware/ dishes) outside the maternal body — succinct one-line explanation expected.
- Variants: IVF (standard), ICSI (single-sperm injection), embryo transfer; basic safety/ethical mention may be required in longer answers.

Kaise Poochha Jaata Hai:

Short 2-mark question: 'Write the basic steps followed in the Assisted Reproductive Technologies (ART) programme... Why is it also known as test tube baby programme?' Expect step-list and the one-line reason (in vitro fertilisation). Longer 5-mark treatments require more detail on hormone stimulation and embryo transfer timing.

Repeat Pattern:

Direct short/long questions on ART steps appear in Paper 4 (Section B) and similar reproductive sections in other papers — moderate-to-high repeat for a compact stepwise answer.

Key Terms:

IVF

oocyte retrieval

embryo transfer

ICSI

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16. Selection of transformed bacteria using antibiotic markers (ampicillin / tetracycline) and insertional inactivation

High Conf. High Priority

Appeared in: Paper 4 · Paper 2 · Paper 3 · 3× total

Question types: MCQ: 1-2 | Short: 1-2 | Long: 0-1 | Case Study: 1

Kya Padhna Hai:

- Core concept: antibiotic resistance genes (e.g., ampicillin resistance, tetracycline resistance) on plasmid vectors are used as selectable markers. After transformation, bacteria grown on antibiotic-containing medium allow only transformants (with plasmid) to survive.
- Insertional inactivation principle: when foreign DNA is inserted into a marker gene (e.g., lacZ), that gene's function is disrupted; colonies can be screened (e.g., blue/white screening using lacZ and X-gal) to identify recombinant clones.
- Plasmid map terms: cloning sites (multiple cloning site/MCS), selectable marker genes, reporter genes (lacZ), and use of antibiotic plates for selection — be ready to identify transformed colony on a plate diagram (as in Paper 4).

Kaise Poochha Jaata Hai:

Questions appear as diagram-based identification ('which colony is transformed? justify'), short definitional items ('what are the sites called where ampicillin and tetracycline resistance genes are inserted? state their role'), or as a single-line role of ;"Öv Æ 7F÷6–F 6R –â –ç6W tional inactivation. Prepare to name colonies and explain selection principle concisely.

Repeat Pattern:

Selection-of-transformants and antibiotic-marker concepts recur across Papers 4 and 2 (diagram and MCQ), and insertional inactivation asked directly as an alternative short question — high recurrence for conceptual and diagram-based testing.

Key Terms:

- *ampicillin resistance*
- *tetracycline resistance*
- *selectable marker*
- *insertional inactivation*
- *lacZ blue-white screening*

17. Trophoblast function in human embryo

Medium Conf. | Medium Priority

Appeared in: Paper 1 · Paper 4 · 2× total

Question types: Short: 2

Kya Padhna Hai:

- Direct asked item: 'Mention the function of trophoblast in the human embryo.'
- Role: implantation and formation of part of placenta (phrased in papers as trophoblast function)
- Secretory role: trophoblast contribution to hormonal signalling (contextual in placenta/implantation Qs)
- Related structure: relation to chorion/placenta formation when asked with placenta hormone questions

Kaise Poochha Jaata Hai:

Very short question asking single function; appears as alternative in Section B (2 marks)

Repeat Pattern:

Seen in Paper 1 and Paper 4 as short 2-mark recall — moderately consistent

Key Terms:

- *trophoblast*
- *implantation*
- *placenta*

18. Tears as immune barrier: type and example

Medium Conf. Medium Priority

Appeared in: Paper 1 · Paper 3 · 2× total

Question types: Short: 2

Kya Padhna Hai:

- Exact question: 'What function is performed by tears shed by our eyes? Which type of barrier is this in immunity? Give another similar example.'
- Function: mechanical washing + lysozyme secretion — name lysozyme as antimicrobial enzyme
- Immunity type: 'innate external barrier' or 'first line of defence' phrasing used in paper
- Provide similar example asked: mucus in respiratory tract or skin surface (paper expects a named example)

Kaise Poochha Jaata Hai:

Very short answer: name function + barrier type + one similar example (2-mark format)

Repeat Pattern:

Short direct Q appears in Paper 1 and multiple papers test 'barrier' examples — moderate repeat

Key Terms:

tears

lysozyme

innate barrier

19. Decomposition pathway to humus (stages: fragmentation, catabolism, mineralisation, humification)

Medium Conf. Medium Priority

Appeared in: Paper 1 · Paper 2 · 2× total

Question types: Short: 2-3

Kya Padhna Hai:

- Exact phrase used: 'Explain this process of decomposition of detritus till the formation of humus.'
- Step order: fragmentation !' leaching !' catabolism !' mineralisation !' humification (Paper 2 lists steps)
- Explain one step: Paper 2 asked specifically to 'explain the fourth step' (humification/mineralisation depending on ordering) — be ready to define and give mechanism
- Agents: role of microbes and detritivores in different steps when asked

Kaise Poochha Jaata Hai:

Short answer asking ordered steps and explanation of a named intermediate step (2 marks)

Repeat Pattern:

Asked as short explanatory Q in Paper 2 and related decomposition mention in Paper 1 — moderate repeat

Key Terms:

fragmentation

humification

mineralisation

20. Wind-pollinated plant adaptations (3 adaptations)

Medium Conf. Medium Priority

Appeared in: Paper 1 · Paper 2 · 2× total

Question types: Short: 3

Kya Padhna Hai:

- Question phrasing: 'Give any three adaptations found in wind-pollinated plants.'
- Expected specifics: reduced/absent petals, large feathery stigmas, light/non-sticky pollen (these are standard phrasing used in paper)
- Example plants: grasses and many trees referenced in similar questions elsewhere in set
- Function mapping: match each adaptation to role in wind pollination as required by 3-mark answer

Kaise Poochha Jaata Hai:

Short 3-point listing with brief function explanation (Section C style)

Repeat Pattern:

Appears in Paper 1 and Paper 2 Section C — moderately repeat

Key Terms:

wind-pollinated

feathery stigma

non-sticky pollen

21. Replication fork: definition and role in replicating long DNA strands

Medium Conf. Medium Priority

Appeared in: Paper 1 · Paper 3 · 2× total

Question types: Short: 3

Kya Padhna Hai:

- Direct question text: 'What is a replication fork? How does it aid replication of long strands of DNA?'
- Definition: replication fork is Y-shaped region where DNA unwinds and replication proceeds bidirectionally
- Mechanism specifics: leading and lagging strand synthesis, Okazaki fragments aiding long-strand replication
- Enzyme mention: DNA helicase and DNA polymerase role often expected in explanation

Kaise Poochha Jaata Hai:

Short 3-mark explanation: define fork + explain bidirectional replication and Okazaki fragments

Repeat Pattern:

Appeared explicitly in Paper 1 and related enzymatic replication questions in other papers — moderate repeat

Key Terms:

replication fork

Okazaki fragments

DNA helicase

22. DNA polymorphism: definition and two applications

Medium Conf. Medium Priority

Appeared in: Paper 1 · Paper 4 · 2× total

Question types: Short: 3

Kya Padhna Hai:

- Question phrasing: 'What is polymorphism in DNA? Give any two applications of DNA polymorphism.'
- Definition: presence of two or more alleles at a locus within population (DNA level variation)
- Applications named in papers: DNA fingerprinting (forensics), paternity testing, population genetics
- Technology tie-in: RFLP, RAPD or microsatellite usage as examples expected in application bullets

Kaise Poochha Jaata Hai:

Short 3-mark question: define + list two applications (named techniques or fields)

Repeat Pattern:

Asked explicitly in Paper 1 and related DNA-fingerprint questions in other papers — moderate repeat

Key Terms:

- DNA polymorphism
- RFLP
- DNA fingerprinting

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23. Pathogen detection at low concentrations: PCR and diseases detected (two examples)

Medium Conf. Medium Priority

Appeared in: Paper 1 · Paper 2 · 2× total

Question types: Short: 3

Kya Padhna Hai:

- Phrase from paper: 'Mention a method that can detect a pathogen at very low concentrations... Which two diseases are detected by this method?'
- Method name: PCR (polymerase chain reaction) — be ready to mention sensitivity and primer-target specificity
- Diseases examples in papers: HIV and Hepatitis C are mentioned in drug-use/HIV tables (Paper 2), or any viral detection example
- Technical note: PCR basis — amplification of target DNA even from low template; real-time PCR as sensitive variant may be mentioned

Kaise Poochha Jaata Hai:

Short 3-mark Q: name method + two disease examples detected by it

Repeat Pattern:

Appears in Paper 1 and Paper 2 — moderate repeat

Key Terms:

- PCR
- HIV
- Hepatitis C

24. Mutation as a source of evolution (how mutations result in evolution)

Medium Conf. | Medium Priority

Appeared in: Paper 1 · Paper 2 · 2× total

Question types: Short: 1-2 | Case Study: 1-2

Kya Padhna Hai:

- Question wording: 'How can mutations result in evolution ?' (Paper 1 choice in case)
- Mechanism: new alleles arise, selection acts on variation, accumulation causes divergence — phrase 'continuous accumulation of changes' used in Paper 2 evolution passage
- Types: point mutations, substitutions leading to new alleles (Paper 2 mentions 'substitution of a new gene for the old one')
- Expected short explanation: mutation !' allele frequency change !' selection/drift !' speciation

Kaise Poochha Jaata Hai:

Case-option short explanation (2 marks) in context of Hardy-Weinberg or evolution passage

Repeat Pattern:

Appears as an internal choice in Paper 1 and as main idea in Paper 2 evolution passage — moderate repeat

Key Terms:

- *mutation*
- *allele frequency*
- *substitution*

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25. Gonadotropin source and ovarian changes during menstrual cycle

Medium Conf. | Medium Priority

Appeared in: Paper 1 · Paper 4 · 2× total

Question types: Long: 5

Kya Padhna Hai:

- Exact phrasing: 'Name the source of gonadotropins in human females. Explain the changes brought about in the ovary by these hormones during menstrual cycle.'
- Source: pituitary (anterior pituitary) secreting FSH and LH
- Ovarian events: follicular development, ovulation (LH surge), corpus luteum formation and degeneration — map to hormonal peaks
- Hormone names & effects: FSH (follicle growth), LH (ovulation + luteinization) — expected in answer

Kaise Poochha Jaata Hai:

Long 5-mark subquestion expecting source + detailed ovarian hormonal/structural changes during cycle

Repeat Pattern:

Appears in Paper 1 and Paper 4 E-section questions on reproduction — moderate repeat

Key Terms:

- *FSH*
- *LH*
- *corpus luteum*

26. Placenta-only hormones produced during pregnancy (name two)

Medium Conf. Medium Priority

Appeared in: Paper 1 · Paper 4 · 2× total

Question types: Short: 1-2 | Long: part of long Q

Kya Padhna Hai:

- Paper phrase: 'Name any two specific hormones which are produced by placenta only during pregnancy.'
- Hormones to name: human chorionic gonadotropin (hCG), human placental lactogen (hPL/placental lactogen), relaxin often produced by placenta/ corpus luteum depending on framing
- Associated function mention sometimes expected (maintenance of corpus luteum by hCG, metabolic effects by hPL)
- Placement: appears as (ii) subpart in a 5-mark reproduction question

Kaise Poochha Jaata Hai:

Short naming with brief function (in subpart of long question)

Repeat Pattern:

Appears across Paper 1 and Paper 4 reproductive clusters — moderate repeat

Key Terms:

hCG

hPL

placenta hormones

27. Stem cells in human embryo: location and significance

Medium Conf. Medium Priority

Appeared in: Paper 1 · Paper 3 · 2× total

Question types: Long: 1-2

Kya Padhna Hai:

- Direct phrase: 'Where are the stem cells located in the human embryo? What is their significance?'
- Location: inner cell mass of blastocyst (embryonic stem cells) and trophoblast (trophectoderm) distinction if needed
- Significance: pluripotency, regenerative medicine, organ/tissue formation — be ready to name pluripotency
- Practical points: embryonic vs adult stem cells distinction when asked elsewhere

Kaise Poochha Jaata Hai:

Short/part of long question (5-mark reproduction cluster) — name location + 1-2 sentence significance

Repeat Pattern:

Appears in Paper 1 and Paper 3 in reproductive biology sections — moderate repeat

Key Terms:

inner cell mass

pluripotent

stem cells

28. Mutualistic interaction and co-evolution example (Ophrys-bee type)

Medium Conf. Medium Priority

Appeared in: Paper 2 · Paper 4 · 2× total

Question types: Short: 2-3 | Long: 1-2

Kya Padhna Hai:

- Paper phrasing: 'Explain with an example how mutualistic interaction involves co-evolution.' and 'How is Ophrys pollinated by a specific species of bee? Describe co-evolution.'
- Example specifics: Ophrys orchid mimics female bee morphology/scent to attract male bees (sexual deception) — mechanism name used in Paper 2/4
- Concept: reciprocal adaptations over time (co-evolution) and pollinator specificity
- Preparation: know both behavioural/ecological and morphological cues used by Ophrys

Kaise Poochha Jaata Hai:

Short/descriptive question: explain example + define co-evolution with justification; sometimes part (a)/(b) two-step 3-mark

Repeat Pattern:

Ophrys-coevolution theme appears across Paper 2 and Paper 4 — medium repeat

Key Terms:

Ophrys

sexual deception

co-evolution

29. Endemism definition and hotspots in India (name two)

Medium Conf. Medium Priority

Appeared in: Paper 1 · Paper 2 · 2× total

Question types: Short: 1-2 | Long: 1

Kya Padhna Hai:

- Exact wording: 'What is endemism ? Name any two hotspots found in India.'
- Definition: species unique to a defined geographic area
- Hotspots examples required: Western Ghats, Eastern Himalaya, Indo-Burma, Sundaland — two expected
- Tie-ins: conservation importance and hotspot criteria sometimes asked in other papers

Kaise Poochha Jaata Hai:

Short 5-mark question part: define + list two hotspots (1-2 sentence each)

Repeat Pattern:

Appears in Paper 1 and Paper 2 in biodiversity sections — moderate repeat

Key Terms:

endemism

Western Ghats

Eastern Himalaya

30. Population density calculation from natality, mortality, immigration, emigration (basic arithmetic problem)

Medium Conf. Medium Priority

Appeared in: Paper 1 · Paper 2 · 2× total

Question types: Long: 1 (calculation)

Kya Padhna Hai:

- Paper phrasing: 'If population density (N) at time (t) is 500, what would the population density be at (t+1) with natality=100, mortality=50, immigration=300, emigration=250?'
- Formula: N(t+1)=N + (births - deaths) + (immigration - emigration)
- Calculation specifics: compute net change (100-50)+(300-250)=50+50=100 !’ N(t+1)=600 as taught
- Be ready to show arithmetic and state final density explicitly

Kaise Poochha Jaata Hai:

Numerical 5-mark problem requiring stepwise calculation and final answer (presented in Paper 1 Section E)

Repeat Pattern:

Computation-style question seen in Paper 1 and similar population questions in Paper 2 — moderate repeat

Key Terms:

- natality
- mortality
- immigration

31. Spermatogenesis hormones: site of FSH action, LH role, and accessory ducts path

Medium Conf. Medium Priority

Appeared in: Paper 3 · Paper 4 · 2× total

Question types: Short: 1-2 | Long: 1-2

Kya Padhna Hai:

- Paper lines: 'State the site of action of FSH in the testes and describe its action' and alternative 'Describe role of LH in spermatogenesis.'
- FSH site/action: acts on Sertoli cells to stimulate spermatogenesis, support spermatid maturation
- LH role: acts on Leydig cells to stimulate testosterone production supporting spermatogenesis
- Accessory ducts: pathway from seminiferous tubules !’ rete testis !’ epididymis
- Cell-stage mapping: which cells undergo mitosis/meiosis as subpart in Paper 3

Kaise Poochha Jaata Hai:

Short 2-mark focused hormonal role question with a small subpart on ducts or cell-stage naming

Repeat Pattern:

Hormonal control of spermatogenesis appears in Paper 3 and Paper 4 case questions — moderate repeat

Key Terms:

- FSH
- Sertoli cells
- Leydig cells

32. Ribozyme identity (23S rRNA) and catalytic rRNA

High Conf. Medium Priority

Appeared in: Paper 2 · 1× total

Question types: MCQ: 1 | Short: 0 | Long: 0 | Case Study: 0

Kya Padhna Hai:

- Fact: In bacteria the ribozyme (catalytic RNA) that acts as a catalyst is 23S rRNA (part of the 50S ribosomal subunit). Memorise the specific 23S rRNA identity as the correct MCQ answer.
- Context: the question tests recognition that some rRNA molecules have catalytic activity (peptidyl transferase) and are ribozymes.
- Linked concept: peptidyl transferase activity of the large ribosomal RNA (not protein) — useful to mention if asked for mechanism.

Kaise Poochha Jaata Hai:

MCQ format: name/choose which rRNA (options: 28S, 5.8S, 26S, 23S) is the bacterial ribozyme acting as catalyst. Expect single-line factual recall.

Repeat Pattern:

Appeared as a single MCQ in Paper 2. High reliability for fact-based MCQ; unlikely to be expanded into long answer in these papers.

Key Terms:

23S rRNA

ribozyme

peptidyl transferase

33. Baculoviruses used as biocontrol: three reasons

Low Conf. Low Priority

Appeared in: Paper 1 · 1× total

Question types: Short: 3

Kya Padhna Hai:

- Exact Q: 'Though baculoviruses are pathogens, they are used as biological control agents. Give three reasons why they are preferred.'
- Typical expected reasons in paper context: host-specificity to target pests, safety to non-target organisms including humans, ability to be mass-produced/formulated
- Application: used as bioinsecticide examples (e.g., Helicoverpa control) when giving reasons
- Formulation note: viral persistence/field acceptability occasionally required

Kaise Poochha Jaata Hai:

Short 3-mark listing question requiring three concrete reasons (as above)

Repeat Pattern:

Seen once (Paper 1) — low repeat, but specific phrasing demands three concrete reasons

34. Insulin structure, maturation and immune reaction to animal insulin

Low Conf. Low Priority

Appeared in: Paper 1 · 1× total

Question types: Long: 1

Kya Padhna Hai:

- Exact paper text: 'Describe the structure of Insulin. How does maturation of Insulin occur? What reaction can occur in a person who is given insulin from an animal source?'
- Structure: A and B chains linked by disulfide bonds, preproinsulin !’ proinsulin !’ mature insulin via cleavage of C-peptide
- Maturation steps: signal peptide removal and C-peptide cleavage by prohormone convertases
- Immune reaction: allergic/hypersensitivity reaction possible on heterologous (animal) insulin administration
- Clinical tie-in: mention C-peptide presence as maturation evidence if needed

Kaise Poochha Jaata Hai:

5-mark long answer requiring structure diagram description, maturation steps and immunological consequence with named reaction

Repeat Pattern:

Observed only in Paper 1 long-section — low repeat

35. Competence of bacterial cell for transformation using calcium ions

Low Conf. Low Priority

Appeared in: Paper 3 · 1× total

Question types: Short: 2-3

Kya Padhna Hai:

- Paper phrasing: 'Why should a cell be made competent to take up an alien DNA ? How can a bacterial cell be made competent using calcium ions? Explain.'
- Reason: competence increases DNA uptake efficiency for transformation (named as technical term)
- Procedure: Ca^{2+} treatment makes cell membrane permeable (cold $CaCl_2$ + heat-shock steps) — mention heat-shock step because paper asks 'how'
- Expected short procedure steps: calcium incubation, heat shock, recovery on medium
- Related terms: plasmid uptake and selection on antibiotic plates (later covered in selection topics)

Kaise Poochha Jaata Hai:

Short 3-mark descriptive procedural answer expected (Paper 3 Section E)

Repeat Pattern:

Specific procedure asked only in Paper 3 — low repeat

Related Topic Pairs

Yeh topics exam mein aksar ek saath poochhe jaate hain — inhe milake padho.

1. Recombinant DNA steps !” Restriction enzyme naming/choice: enzyme selection and compatible ends are repeatedly tested together (Paper 2 flow diagram + enzyme-choice question).

2. Activated sludge in STP !” Anaerobic sludge digestion: both are asked in same short question set (Paper 2) and in STP flow questions (Paper 4), so study together.

3. Vaccine long-term immunity !” Memory cells/anamnestic response: case questions ask both property name and cellular basis together (Paper 1).

4. Seed advantages !” Perisperm vs Pericarp !” Polyembryony: these three subparts are grouped in the same long question (Paper 1 Section E).

Apni Strategy Chuno

Pick your goal — here's exactly which topics to study based on past paper patterns.

Bas Pass Hona Hai

Just want to pass — 16 of 35 topics · ~58% marks coverage

1. Microspore formation: meiosis of pollen mother cell
2. Ploidy of primary and secondary spermatocytes (spermatogenesis)
3. Polygenic inheritance: deviation from Mendelian patterns
4. Non-Mendelian inheritance in human blood groups (multiple allelism & co-dominance)
5. Methods to introduce alien DNA into plant and animal cells (e.g., microinjection, Agrobacterium)
6. Logistic population growth model (Verhulst-Pearl) and equation (K, r)
7. Uterus wall structure: layers and functions (endometrium, myometrium, perimetrium)
8. Predator vs Prey: distinction and two ecological roles of predators
9. Hardy-Weinberg equilibrium: definition and factors (founder effect, gene migration)
10. Long-term immunity from vaccination: name, property (memory), types and memory cells/anamnestic response
11. Angiosperm seed advantages and features: seed vs perisperm vs pericarp and polyembryony example
12. Activated sludge in Sewage Treatment Plant (STP) and anaerobic sludge digestion
13. Recombinant DNA technology steps: cloning, transfer, replication and term 'replication of recombinant DNA' (technical term for Step 4)
14. Restriction endonuclease naming convention and recognition site length (e.g., EcoRI, HindII, 4/6 base cutters)
15. Selection of transformed bacteria using antibiotic markers (ampicillin / tetracycline) and insertional inactivation
16. Assisted Reproductive Technologies (ART) steps (test-tube baby)

Average Score Chahiye

Decent score — 32 of 35 topics · ~96% marks coverage

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17. Trophoblast function in human embryo
18. Tears as immune barrier: type and example
19. Decomposition pathway to humus (stages: fragmentation, catabolism, mineralisation, humification)
20. Wind-pollinated plant adaptations (3 adaptations)
21. Replication fork: definition and role in replicating long DNA strands
22. DNA polymorphism: definition and two applications
23. Pathogen detection at low concentrations: PCR and diseases detected (two examples)
24. Mutation as a source of evolution (how mutations result in evolution)
25. Gonadotropin source and ovarian changes during menstrual cycle
26. Placenta-only hormones produced during pregnancy (name two)
27. Stem cells in human embryo: location and significance
28. Mutualistic interaction and co-evolution example (Ophrys-bee type)
29. Endemism definition and hotspots in India (name two)
30. Population density calculation from natality, mortality, immigration, emigration (basic arithmetic problem)
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32. Ribozyme identity (23S rRNA) and catalytic rRNA

🎯 Top Karna Hai

Full coverage — all 35 topics · 100% marks coverage

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33. Baculoviruses used as biocontrol: three reasons
34. Insulin structure, maturation and immune reaction to animal insulin
35. Competence of bacterial cell for transformation using calcium ions

Overall Exam Strategy

Bas Pass Hona Hai: focus on high-frequency factual items (ribozyme identity, recombinant DNA steps/ selection markers, STP activated sludge and aeration tank role, lac operon induction, ART steps, gel electrophoresis principle/EtBr, double fertilization sequence). Practice short definitions + 1–2 sentence mechanisms and one diagram each (replication fork, uterus/ovary changes, gel lanes/selection plate).